

Math 526, S16

Name: *Answer*

Quiz 10

Section:

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Find the determinant of A where $A =$

$$\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 4 & 5 \\ 5 & 4 & 0 & 3 \\ 2 & 0 & 0 & 1 \end{bmatrix}$$

using the

co-factor expansion.

$$\det A = \begin{vmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 4 & 5 \\ 5 & 4 & 0 & 3 \\ 2 & 0 & 0 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 3 & 4 & 5 \\ 4 & 0 & 3 \\ 0 & 0 & 1 \end{vmatrix} - 2 \begin{vmatrix} 0 & 3 & 4 \\ 5 & 4 & 0 \\ 2 & 0 & 0 \end{vmatrix}$$

$$= 1 \cdot 1 \cdot \begin{vmatrix} 3 & 4 \\ 4 & 0 \end{vmatrix} - 2 \cdot 2 \begin{vmatrix} 3 & 4 \\ 4 & 0 \end{vmatrix}$$

$$= (3 \times 0 - 4 \times 4) - 4(3 \times 0 - 4 \times 4)$$

$$= -16 + 64$$

$$= 48$$

Problem 2. (4 points) True or false (give a reason if true and a counter example if false)

(a) The determinant of $A - B$ equals $\det A - \det B$.

False.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \quad A - B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\det A = 1 \quad \det B = 4$$

$$\det(A - B) = 1 \neq -3 = \det A - \det B$$

(b) AB and BA have the same determinant.

True

$$\det(AB) = \det A \det B = \det B \det A = \det(BA)$$